

Assignment 1A

Group 5
Societal Problem Definition



ALJAZEERA

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Nepal-Tibet floods: What happened, what caused them and who is missing?

More than 390 killed, 1,500 missing after a glacial collapse triggered catastrophic floods in Nepal and Tibet.

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Glacier collapse unleashes catastrophic downstream flooding

Melimopoulos, Elizabeth. "Nepal Floods: What Happened, What Caused Them and Who Is Missing?" *Al Jazeera*,
www.aljazeera.com/news/2026/8/27/nepal-tibet-floods-what-happened-what-caused-them-and-who-is-missing.

Task 1

The news article titled “Nepal Floods: What Happened, What Caused Them and Who Is Missing?” was written by Elizabeth Melimopoulos, writer from Al Jazeera (a Qatari based international news outlet). Al Jazeera had obtained the information about the outbreak from Reuters (another media outlet based in the UK). The article gives an overview of the recent Tibet-Nepal floods in 2026. The overview contains the floods' causes, geographical impact, casualties, missing people, rescue efforts, and the possible role of climate change.

Task 2

The main challenge is responding quickly and effectively to floods when roads, bridges and communication infrastructure are destroyed. Authorities would need accurate information about affected people and locations to help with the rescue.

We are interested because natural disasters can affect many people in remote areas, so better organisation of information could help emergency services respond more efficiently.

This natural disaster is especially concerning for the following people:

- Self-reliant farmers - their food source and entire livelihood is gone
- People working at hydropower plants - a lot of workers lost their lives or got trapped
- Families which had their houses alongside the river.
- Nepalese government and local authorities – coordinate disaster response.
- Hospitals and healthcare workers – treat affected people.
- Emergency services - they risk their lives in high-altitude rescues
- NGOs and humanitarian organisations – provide food, supplies and aid.

Task 3

Article name:

"Societal impacts and vulnerability to floods in Bangladesh and Nepal"

Relevance to the challenge:

The article is relevant to the data management problem in Nepal during floods, as network stations are destroyed and data collection becomes harder or impossible. Consequently, a lot of critical information goes missing, making it very hard to rescue people.

Subtype of scientific article:

Review.

Citation:

Dewan, T. H. (2015). Societal impacts and vulnerability to floods in Bangladesh and Nepal. *Weather and Climate Extremes*, 7, 36–42.

<https://doi.org/10.1016/j.wace.2014.11.001>

Article name:

“An institutional analysis of glacial floods and disaster risk management in the Nepal Himalaya”

Relevance to the challenge:

This article is significant because it talks about how Nepalese communities manage glacial flood risks. It highlights problems such as poor communication between organisations, limited sharing of scientific information, and insufficient coordination. Therefore, it is related to our challenge of improving the organisation and sharing of information during flood disasters.

Subtype of scientific article:

Original research article: The study uses a mixed approach, combining household surveys with qualitative analysis.

Citation:

Thompson, I., Shrestha, M., Chhetri, N., & Agusdinata, D. B. (2020). An institutional analysis of glacial floods and disaster risk management in the Nepal Himalaya.

International Journal of Disaster Risk Reduction, 47, 101567.

<https://doi.org/10.1016/j.ijdrr.2020.101567>

Article name:

“Harnessing trustable crowdsourcing power for flood disaster evaluation”

Relevance to the challenge:

This article is relevant, because it addresses the problem by proposing a system based on information and communication technology. It utilizes crowdsourcing to

gather and distribute precise, real-time data to regions impacted by flooding. The system keeps this crowdsourced information reliable by flagging and filtering out submissions from unreliable and malicious contributors. The paper also introduces an user reputation score algorithm to identify trusted users. By getting relevant information from flooded areas, the system can help relief organizations respond quicker and more effectively to floods.

Citation:

Abassi, S., Vahdat-Nejad, H., & Moradi, H. (2024). Harnessing trustable crowdsourcing power for flood disaster evaluation. *Natural Hazards*, 120:8723–8741. <https://doi.org/10.1007/s11069-024-06547-8>

Subtype of scientific article: Original research article (it proposes a system and validates it through simulation, rather than reviewing or offering an opinion)